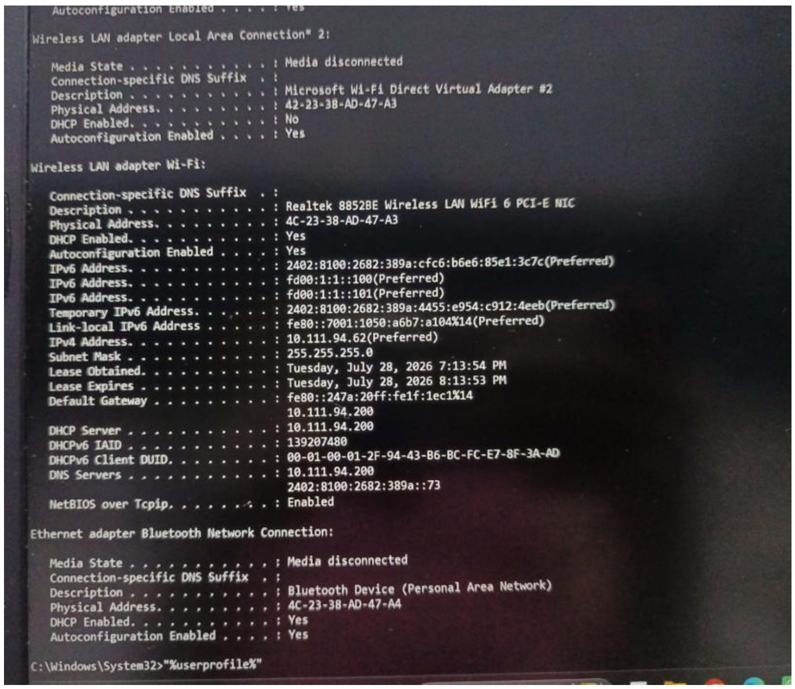
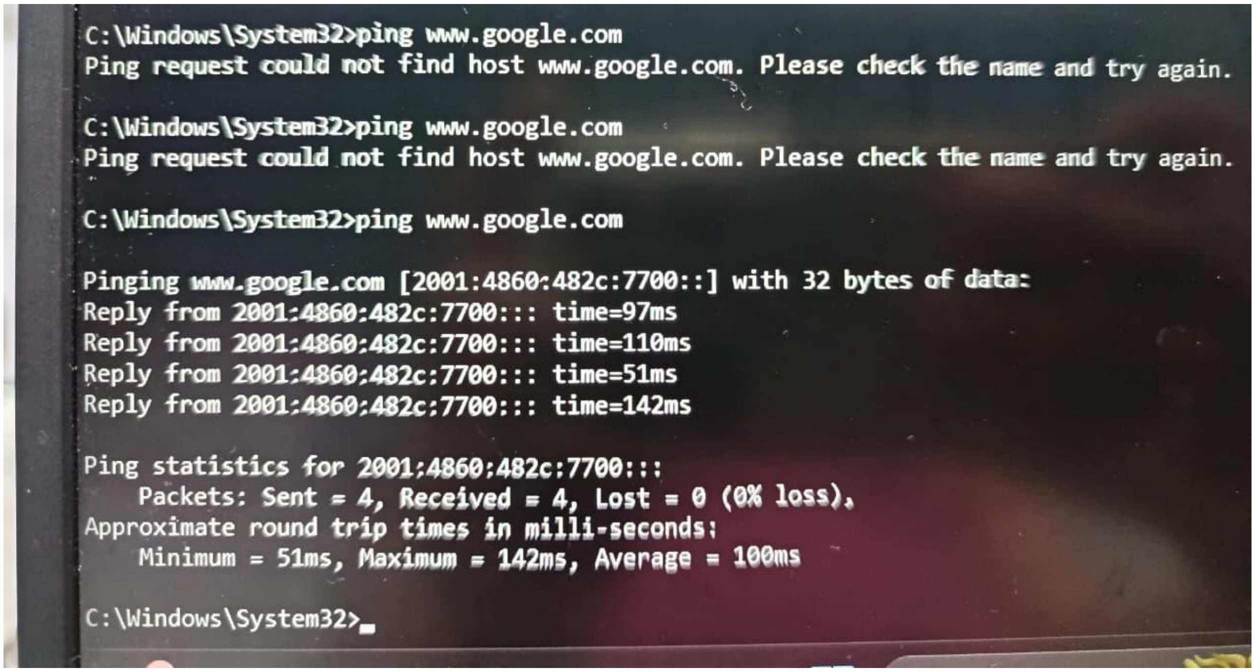
NETWORK TROUBLESHOOTING, MANAGEMENT , MOINTORING AND OPTIMIZATION

1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



Parameter	Your Value
IPv4 Address	10.111.94.62
Subnet Mask	255.255.255.0
Default Gateway	10.111.94.200
DNS Server(s)	10.111.94.200

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.



Short Note

- Disconnected: Ping failed because there was no network connection.
- Reconnected: Ping succeeded, showing the computer could communicate with Google's server. This means the Internet connection was restored

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Steps

1. Press Ctrl + Shift + Esc to open Task Manager.
2. Click the Processes tab.
3. Play a YouTube video for 5 minutes.

4. Observe the Network column.

Example Result

Process	Peak Network Usage
Google Chrome	2.9 Mbps

**4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.**

Steps

1. Install and open Wireshark.
2. Start capturing on your Wi-Fi or Ethernet interface.
3. Browse Flipkart or Instagram Web for 30 seconds.
4. Stop the capture.
5. In the filter bar, type:
    - http (for HTTP traffic)
    - tls (recommended for HTTPS traffic)

Three Types of Information You Can See

Information	**Description**
Source IP Address	IP address of the sender.
Destination IP Address	IP address of the receiver/server.
Protocol Details	HTTP request, TLS handshake, packet size, etc.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first

Home Network Speed & Reliability Checklist

- Restart the modem and Wi-Fi router.
- Move closer to the Wi-Fi router or use a Wi-Fi extender.
- Use an Ethernet cable instead of Wi-Fi for gaming.
- Disconnect unused devices from the network.
- Close background downloads, updates, and streaming apps.
- Update the router firmware.
- Restart the gaming PC or console before playing.

Suggestion I Would Try First

Use an Ethernet cable instead of Wi-Fi.

Reason: A wired connection provides a more stable connection, lower latency (ping), and fewer interruptions during online gaming.